A Design Framework for Virtual Space based on Spectral Relationships

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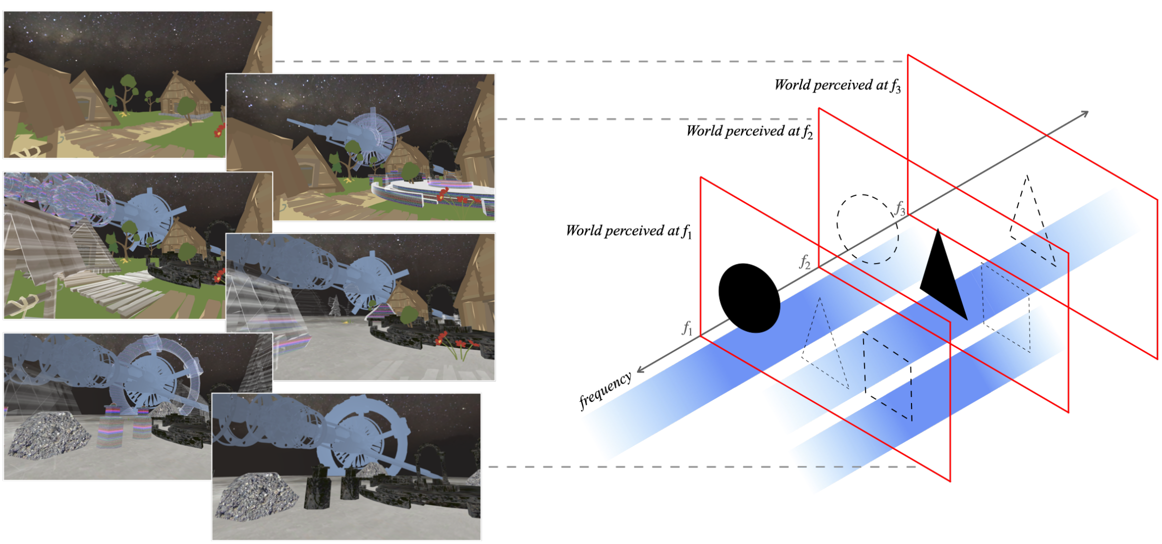

Figure 1: Overview of Spectral Space design. The alpha value of the blue strips stands for the visibility of each entity. Continuous virtual spaces form along the additional non-spatial axis (frequency) through spectral relationships between entities.

Abstract

Conventional Virtual Reality (VR) systems are structured around discrete, fixed worlds, requiring mechanisms such as portals or scene-blending to transition between worlds. This structure limits the creation of subjective, continuous spatial experiences. We introduce Spectral Space, a framework that augments the conventional 3-dimension virtual space with an additional non-spatial axis that we call *frequency*. Virtual entities are assigned with perceptual intensity functions along this axis, allowing their visibility to change smoothly as users adjust their position on the *frequency* axis. This enables multiple worlds to coexist in place and produces intermediate spaces that emerge from the graded overlap of entities. We implemented a prototype system and conducted a user study comparing spectral transitions with two representative VR

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transition techniques (discrete teleportation and continuous dissolve). Results show that Spectral Space can maintain presence and self-location while significantly increasing usable spatial range, promoting exploratory behavior, and supporting layered interpretation of objects and worlds. These findings demonstrate that organizing virtual worlds as continuous spectral layers provides a viable and expressive alternative to scene-based VR, expanding the design space for multi-world interaction and navigation. This work opens new directions for space design strategies and augmenting human experience by redefining the spatial foundation of virtual space.

CCS Concepts

• **Human-centered computing → Virtual reality**; *Mixed / augmented reality*; *Interaction techniques.*

Keywords

Multi-layered Virtual Reality, Virtual Space Design, Spatial Awareness, Transition Technique

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1 INTRODUCTION

Virtual reality (VR) systems are conventionally organized as collections of discrete, self-contained worlds, each with its own geometry and narrative state. When systems incorporate multiple worlds, users transition between them using techniques such as portals and spatial gateways[53], scene-morphing or spatial-alignment methods[54, 55], or cut-scene–based teleportation and fade effects [13, 22, 32, 39]. Starting from one place at a time, recent developments in multi-presence and mixed reality (MR) research demonstrate that user experience does not necessarily need to be confined to a single environment. Multi-presence systems show how users may distribute attention across overlapping virtual spaces or inhabit multiple environments simultaneously[1, 2, 43, 44], while work along the Reality-Virtuality Continuum (RVC)[34] has explored graded transitions between physical and virtual layers[11, 12, 14, 35, 37]. These lines of research point toward more flexible and subjective forms of spatial experience in VR that extend beyond traditional single-fixed-world models.

Across these approaches, a general pattern emerges: while aiming for a fluid and dynamic spatial experience, techniques for transitioning across worlds [13, 22, 54, 55], supporting multi-presence [2, 25, 35, 43, 44, 52], or enabling blended realities [11, 12, 14, 34, 37] typically operate on architectures that treat environments as separable units. This structural rigidity also reflects the fundamental design philosophy of conventional VR, which aims to render a preconstructed world to the user with maximum certainty, leaving little room for subjective spatial interpretation unlike traditional media such as literature or cinema [18]. Consequently, a fluid spatial experience often depends on added mechanisms to coordinate or merge distinct layers. This reflects not a limitation of these techniques, but the historical organization of VR frameworks as a powerful tool to extend humans' spatial experience for discrete, pre-defined worlds, rather than generating worlds based on the user's perspective.

In this work, we propose a different line of thinking:

- What if virtual space were fundamentally continuous rather than assembled from discrete scenes?
- What if virtual space were generated using a logic that incorporates the user's perspective, rather than rendering a fixed, pre-constructed world?
- And within such a structure, could users naturally explore, interpret, and inhabit such virtual space?

To investigate these questions, we introduce *Spectral Space*, a framework that augments the conventional three-dimensional VR coordinate system with an additional non-spatial *frequency axis*. Here, *frequency* is used as a design metaphor to describe a continuous, non-spatial control dimension through which different virtual entities are gradually emphasized or deemphasized, rather than as a physical or signal-based quantity. We map virtual entities onto perceptual intensity functions along this axis, determining how strongly they appear relative to the user's spectral state. In this way, the continuity of the virtual worlds no longer depends on spatial boundaries, but instead emerges naturally from the overlap and gradual transitions of these intensity functions. As users adjust their frequency, entities fade in, fade out, overlap, or partially coexist, producing continuous transitions across multiple virtual worlds. By shifting their position along the frequency axis, users effectively navigate these coexisting layers, revealing intermediate spaces that emerge from the spectral overlap of entities and are difficult to realize in conventional discrete-world systems.

To evaluate this framework, we conducted a user study comparing transition in our spectral space design with two representative conventional transition techniques inside VR (discrete and continuous). The results show that although this interaction method is unconventional, it provides a level of presence comparable to traditional approaches, while significantly increasing the usable spatial range, enhancing exploratory engagement and supporting a novel form of interaction grounded in spectral relationships.

The contributions of this paper are three-fold: (1) We propose a conceptual framework for constructing multilayer continuous virtual spaces based on spectral relationships between entities, introducing frequency as a non-spatial dimension of presence. (2) We realize this framework through system components including a frequency control module, object filtering mechanisms, and spatial layer generation, enabling users to smoothly navigate among multiple coexisting worlds. (3) We validate the approach through a user study comparing it with conventional transition techniques, revealing how spectral space influences users' spatial cognition and exploratory behavior.

2 RELATED WORK

2.1 Challenge of Continuity in Multi-Space VR

Conventional VR and XR systems rely on explicit transition techniques to navigate between multiple environments. These transitions may occur between virtual worlds or across virtual, augmented, and physical spaces along the RVC [34]. Existing approaches can be broadly grouped into three categories: parallel translation, direct-scene transitions, and object-based transitions.

Parallel translation techniques arrange multiple worlds on a conceptual spatial plane and allow users to move between them via locomotion. Portals [53], corridors, and hub-based layouts connect discrete spaces while preserving spatial orientation [40]. Similar strategies extend to mixed reality, where virtual gateways bridge physical and virtual environments [8, 37, 54]. More recent systems such as SceneFusion [55] visually align and stitch environments into a shared coordinate frame, producing blended regions that users can walk through. Despite their emphasis on perceptual continuity, these approaches still restrict users to occupying one discrete world at a time.

Direct-scene transitions replace one environment with another without simulating spatial adjacency. Techniques such as cuts, fades, and dissolves regulate visual change to reduce abruptness [13, 22]. Overlap-based systems such as OVRlap visually superimpose multiple environments, allowing elements from different worlds to coexist within the user's view. While this overlap produces a smoother perceptual transition, it functions primarily as a visual interpolation between predefined scenes rather than a reconfiguration of space itself. The user's self-location effectively jumps between worlds, even when visual blending is continuous.

In mixed-reality systems, similar logic is extended along the RVC by gradually adjusting the balance between physical and virtual elements [3, 36]. These approaches generate continuity by modulating the degree of visibility or dominance of each world, but the underlying environments remain fixed and discretely defined.

A third line of work explores **object-based transitions**, shifting continuity from the scene level to individual entities. Systems such as SelectVisAR [12] and Blending Spaces [11] independently control object visibility, transfer, or representation across realities. VRception [16] further positions objects along a continuous representational spectrum between predefined worlds. Although these approaches allow objects to occupy intermediate states, they still operate over a set of discrete environments and primarily manipulate appearance rather than spatial organization.

Overall, existing methods mainly aim to impose continuity onto fundamentally discrete spaces. Continuity is treated as a transitional effect, achieved through visual blending or object interpolation, rather than as a native spatial property. As a result, intermediate states often remain liminal or non-functional, and the spatial structure itself is not reconfigured as continuous. This motivates approaches that treat continuity as an intrinsic dimension of space, enabling new, usable spatial configurations to emerge rather than merely smoothing transitions between predefined worlds.

2.2 Subjective Spatial Experience in Mediated Space

Spatial experience in mediated environments is not given, but constructed—shaped by how each medium frames, renders, and invites users to imagine space. In literary narratives, space exists almost entirely through the reader's imagination [23]. Painting similarly constructs spatial experience on a two-dimensional surface through perspective, scale, and composition, ranging from realism to abstraction. Theatre occupies a hybrid position: the stage is a physical place, yet its spatial meaning is completed subjectively by the audience [15]. Across these media, spatial experience is inherently subjective, relying on the viewer's active interpretation rather than a fixed spatial reality [18].

As media become more visually constructed, spatial experience becomes increasingly constrained. Cinema presents a pre-authored audiovisual space with deliberate spatial organization [7]. While viewers still interpret meaning within these frames [28], the spatial structure itself is largely fixed by the creator. Interactive digital games extend this further by combining the player's physical space, the virtual game world, and an abstract operational space [26]. Although the environment is predesigned, players actively construct spatial meaning through interaction and perspective.

Virtual reality takes immersion to an extreme. In conventional VR systems, achieving a stable and objective spatial experience is often treated as a primary goal. Presence—the subjective feeling of "being there" [21, 30, 42, 46, 47, 49]—is widely used as a core evaluation metric, alongside related constructs such as co-presence [6, 33], spatial presence [5, 20, 56], and agency [20, 24]. Achieving high presence typically requires the interface to become transparent [50], minimizing subjective divergence and treating breaks in presence as failures [14, 48]. As a result, conventional VR often assumes that all users should experience the same predefined space as consistently as possible.

Recent work, however, suggests that people are looking for a more fluid and flexible spatial experience in VR. One line of research explores multi-presence, the experience of being present in multiple virtual environments simultaneously [2, 43, 44, 52]. Systems such as OVRlap [44] enable users to perceive and act across overlapping virtual environments by superimposing multiple viewpoints, while dynamically shifting attentional focus. Related XR systems support layered spatial experiences by allowing physical and virtual elements to coexist along the Reality–Virtuality Continuum [11, 12, 31, 43]. These approaches introduce fluidity at the perceptual level, yet retain fixed underlying spatial structures.

Other systems adapt the virtual environment in response to the user's internal state. Wizard of VR [17] alters environmental properties based on physiological signals such as EEG and heart rate variability, while Emotional Beasts [4] generates expressive avatars from users' emotional states. In multi-user contexts, systems such as MultiplexedVR [27] and XRLive [31] allow participants to inhabit different environments while maintaining mutual awareness and communication.

Together, these works demonstrate that users can maintain presence within fluid, overlapping, and personalized spatial experiences. However, subjectivity in these systems is largely achieved through surface-level modulation: layering, blending, or adaptation applied to fundamentally fixed, predesigned worlds. When multiple users are involved, this often introduces trade-offs between individual perspectives and shared spatial consistency. Unlike traditional media, where subjective spatial construction is intrinsic, most VR systems lack a spatial foundation that supports restructuring or generation of space in response to the user's perspective. Addressing this gap requires rethinking spatial continuity itself as a native property of virtual environments, balancing subjective experience with shared spatial coherence.

2.3 Preparation Workload for Overlaying Different Worlds

Although virtual space itself is unconstrained, many systems colocate multiple environments within a shared physical coordinate system to support natural walking, which is known to benefit presence and spatial understanding. Techniques such as Impossible Spaces [51] and Foldable Spaces[19] rely on self-overlapping or folded geometries to maximize walkable area within limited physical space.

However, overlapping architectures exhibit strong geometric dependencies: virtual layouts must be carefully designed to align with one another, often requiring matched boundaries or structural correspondences. This results in a high preparation workload, as environments must be adapted in advance to ensure compatibility. Moreover, perceptual errors in overlapping regions can arise due to orientation changes [41].

Similar constraints appear in approaches that blend environments along the Reality–Virtuality Continuum (RVC) [11, 12, 16, 38]. For example, Pointecker et al. [38] rely on predefined transitional representations to ensure smooth perceptual continuity, requiring explicit preparation of environment elements across multiple states.

Together, these approaches achieve continuity through carefully authored overlaps and transitions, but scale poorly to dynamic or heterogeneous environments. This highlights the need for low workload methods that enable multiple worlds to coexist without extensive geometric or representational preconfiguration.

3 DESIGN RATIONALE

Based on the limitations identified in prior work, we derive three design rationales that guide our system design.

Continuous Spatial Experience. Rather than treating continuity as a transitional or perceptual effect between discrete environments, continuity should be a native property of the virtual space itself. We address this by introducing a non-spatial axis that enables a spectral structuring of environments. This allows multiple worlds to coexist and overlap smoothly, supporting gradual transitions while keeping users perceptually anchored within the space, and enabling additional usable spaces to emerge between worlds.

User-Based Generation and Consistency. The system should support individualized and subjective spatial experiences while maintaining a coherent shared space for multi-user interaction. To this end, we adopt a user-driven perceptual modulation approach in which each user experiences a personalized spatial configuration, while a shared underlying structure ensures coherence and mutual intelligibility across users.

Versatility. We aim for a flexible and scalable method for composing and overlaying multiple environments that does not rely on specific geometries or predefined representations. Our method uses a single-parameter mechanism applied to entities that allows virtual worlds to be overlaid within a shared coordinate system regardless of differences in scale, layout, or object composition.

4 SYSTEM DESIGN

We developed *Spectral Space*, a unique framework for the spectral transition of VR environments. Based on this framework, we implemented a multi-layered VR system in which multiple environments are superimposed and continuously reconfigured according to the user's state. In this section, we describe the framework and its specific implementation procedure.

4.1 Framework

Spectral Space expands the conventional three-dimensional VR coordinate space by introducing an additional non-spatial axis that describes each entity's presentation characteristics as a *perceptual intensity function*. The framework consists of two core operations.

(1) Spectral Structuring of the Environment.

- Each entity possesses its own *perceptual intensity function* on a one-dimensional, non-spatial spectral axis.
- This function specifies how much sensory information the entity provides to the user as a function of the user's spectral state: it reaches its maximum within a certain interval and decays continuously outside that interval.*

* In conventional VR environments, this intensity is implicitly fixed at 1 (i.e., entities are always fully visible). Our framework makes this value explicit and generalizes it into a tunable function for multi-layered perceptual structuring.

(2) User-driven Perceptual Modulation.

- The user possesses their own perceptual intensity function on the spectral axis, and this function is able to be shifted along the axis.
- The perceptual information the user receives from each entity is computed algorithmically based on the relationship between the user's function and the entity's function, and it changes continuously as the user adjusts their spectral position.
- Each user's intensity function is not used for self-perception, but for mutual perception: other users are treated in the same manner as entities, and the strength of their perceptual presence is computed through the same process.

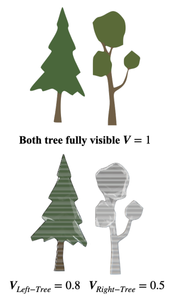
Through these operations, users can continuously modulate the perceptual content of the environment without any physical locomotion, simply by adjusting their position on the spectral axis. Entities are all overlaid within the same three-dimensional coordinate space; however, the environment is structured not only by spatial relationships but also by the collective configuration of perceptual intensity functions. As a result, the VR environment no longer behaves as a fixed three-dimensional structure but is dynamically reconfigured according to the changing perceptual intensities between the user and surrounding entities.

4.2 Implementation

In this study, we integrated the proposed framework into multiple existing VR environments and implemented it by superimposing them along the spectral axis. This section describes the structure of the resulting environment and the mechanism of user interaction, with reference to the schematic illustration shown in Fig.2, Fig.3 and Fig.4.

(1) Spectral Structuring of the Environment. In our implementation, the spectral axis is realized as a frequency parameter f (Hz), inspired by the metaphor of a radio tuning dial and independent

$$V_n(x) = \begin{cases} 1 & x \in [f_{n2}, f_{n3}] \\ \in (0,1) & x \in (f_{n1}, f_{n2}) \cup (f_{n3}, f_{n4}) \\ 0 & x \in (-\infty, f_{n1}] \cup [f_{n4}, \infty) \end{cases}$$



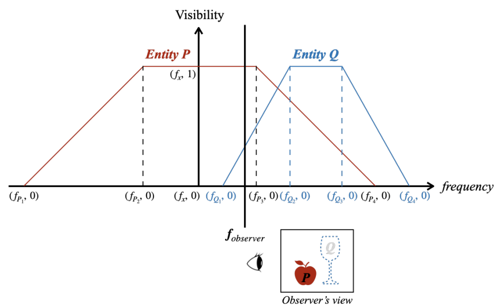


Figure 2: Perceptual intensity and entity visibility. Top: the intensity function $V_n(x)$ and an example entity visibility hierarchy. Bottom: how visibility is applied to compute perceptual intensity.

of any physical quantity. In this system, every object is assigned a trapezoidal perceptual intensity function representing its visual perceptual intensity on the spectral axis. This visibility function is defined by four control points $(f_{n_1}, f_{n_2}, f_{n_3}, f_{n_4})$: objects are rendered in a fully visible state ($v = 1$) within the central interval, invisible ($v = 0$) outside the outer interval, and attenuated ($0 < v < 1$) in between through linear interpolation. The perceptual intensity function, here understood as the visibility function $V_n(x)$ of entity f_n, would be as shown in Fig. 2.

The control points of entities are not placed completely at random. We distributed the entities based on three spectral regions—full-visibility zone, mixed zone, and outer attenuation zone, so that structural consistency of each environment was preserved while giving variability among entities, as shown in Fig. 3. Entities' control points were assigned with local randomness within these regions. We denote the visibility of entity Ni as V_{Ni}:

- Full-visibility zone: all objects appear fully visible. $V_{Ni} = 1$.
- Mixed zone: fully visible, attenuated, and invisible states coexist. $V_{Ni} = 1 \; or \; 0 < V_{Ni} < 1$
- Outer attenuation zone: only attenuated and invisible states appear. That is, no objects are fully visible, $V_{Ni} = 0 \; or \; 0 < V_{Ni} < 1$

By stacking multiple VR environments equipped with this spectral-layer structure along the frequency axis, the system produces a composite environment in which different layers fade in and out continuously as the user's spectral position changes.

(2) User-driven Perceptual Modulation. In our implementation, the center of mass of the user's perceptual intensity function is treated as their spectral state value f_{user}. Adjusting this value, meaning shifting their intensity function along the spectral axis, changes the portion of the multi-layered world that becomes perceptually available to the user. For each frame, the system evaluates every object's visibility function $V_n(x)$ at f_{user}, determining whether the object should appear fully visible, partially attenuated, or invisible. The visibility of all entities is updated continuously as the user moves along the spectral axis.

In Unity, we implement the visibility through a material shader. The value of the visibility function $V_n(x)$ is expressed through two visual components: *alpha transparency* and *the amount of visual noise*. Both components change together according to $V_n(x)$. When $V_n(x) = 1$, the entity is shown clearly with full opacity and no noise. When $0 < V_n(x) < 1$, the shader gradually reduces opacity and increases noise. When $V_n(x) = 0$, the entity becomes fully invisible. In this way, the shader provides a continuous visual transition that reflects the shape of the trapezoidal visibility function defined on the spectral axis.

To allow the user to control f_{user}, we built a dial controller inspired by radio frequency tuning. The device utilized a 360-degree rotary encoder connected to an Arduino Uno R4 microcontroller, which communicated with the VR system, as shown in Fig. 4. The system was programmed to map the angular velocity of the physical dial to the virtual transition speed; specifically, rapid rotation triggered a fast shift in frequency value, while slower rotation resulted in a gradual shift.

5 USER STUDY

5.1 Participants

Participants were 15 (3 identified as female, 12 as male) students at a university, aged 20–31 years ($\mu = 26.7$, $\sigma = 3.02$). 2 participants reported that they were first-time VR users, 6 reported that they had used VR before, and 7 reported that they were regular VR users. All participants had normal or corrected-to-normal vision, and had no motor impairments. All participants gave informed consent to take part in the study. The project was approved by the ethics committee. Questionnaire responses and interviews were analyzed for all participants. However, behavior data from two participants were excluded due to technical issues, leaving a final sample of 13 for the behavioral analysis.

5.2 Apparatus

Participants experienced the virtual environment using a Meta Quest 3 head-mounted display (HMD) (Meta Platforms Inc., Menlo Park, CA, USA), refresh rate 72Hz. The experimental application was developed using Unity (Unity Technologies, San Francisco, CA, USA). The virtual environment consisted of 4 distinct worlds designed to represent different phases along a continuous spectrum. This spectrum axis was defined with a value range from 0 to 10,000, determining the visual characteristics of the surroundings. User input was managed through a hybrid setup involving the custom dial controller for transition and the right Meta Quest 3 Touch Plus controller for locomotion in VR.

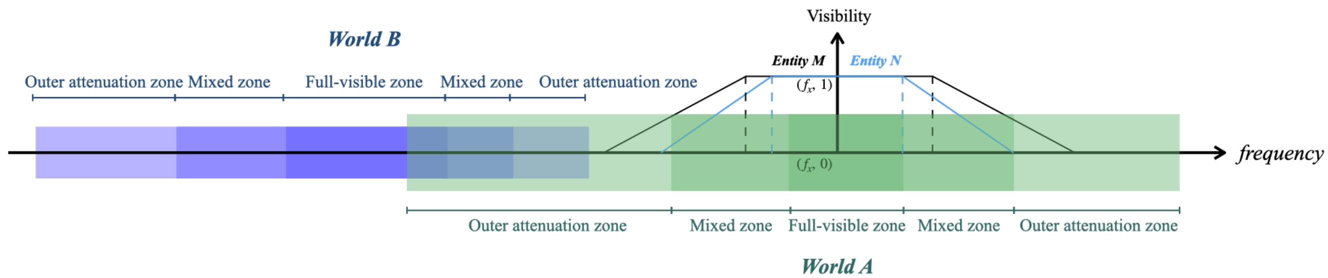


Figure 3: World composition: entity distribution on the non-spatial axis forms continuous worlds

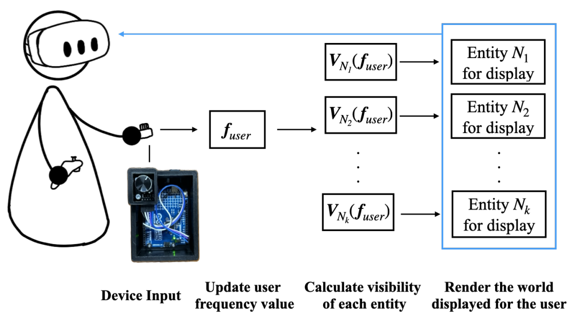


Figure 4: System Architecture: f_{user} stands for user's frequency, and $V_{N_k}(x)$ stands for the visibility function of entity N_k.

5.3 Experimental Design

To investigate how the transition based on the proposed Spectral Space affects users' exploration experience and perception in VR environments, we conducted a user study. We employed a within-subjects design, evaluating participants' experiences across three different transition conditions (*Fade*, *Dissolve*, and *Spectrum*) through quantitative behavioral data, questionnaires and qualitative feedback from interviews.

5.3.1 Task. To facilitate and observe users' exploratory behavior within each condition, we designed a photography task. Participants were assigned the role of a "journalist exploring an unknown virtual world" and were instructed to take exactly four photographs by that they felt best captured the characteristics of the environment. The photo is taken by pressing the index trigger on the right controller. No restrictions were placed on the timing or location of the shots; participants were free to move and explore the space at their own pace, capturing any moments or locations they found meaningful. Crucially, this task was not intended to evaluate the quality or performance of the photographs. Instead, it was designed as a probe to assess the influence of each transition method on users' exploratory behaviors and subjective experiences as they navigated the VR space.

5.3.2 Conditions. To objectively evaluate the characteristics of the proposed *Spectrum* method, we selected two established VR scene transition techniques, *Fade* and *Dissolve*, as comparative baselines. All three conditions used the custom dial controller for transition control. All conditions start from the same visualised world, corresponding to 2000Hz for *Spectrum* condition. The details of these three conditions are as follows:

Fade. This condition serves as a baseline for discrete and discontinuous transitions. When the dial is turned by one increment, the current view fades to black, remains dark for a 0.5-second interval, and then fades in again as the user is instantly relocated to the subsequent scene.

Dissolve. This condition serves as a baseline for continuous and gradual transitions. It employs a cross-fade technique where the current scene fades out while the next scene simultaneously fades in by changing alpha of the two scenes, creating a smooth connection between the two environments. Users can manually control the speed and progress of the transition using the dial, allowing them to pause at any intermediate stage where the two scenes visually overlap. However, this overlap is a uniform scene-level blend rather than a structured or object-level coexistence as in the *Spectrum* condition.

Spectrum. This method utilizes the system implementation detailed in the previous section. Users navigate by adjusting their own frequency parameter. While participants access the same set of worlds presented in the *Fade* and *Dissolve* conditions, the emergence and disappearance of objects within these worlds are dynamically constructed through spectral superposition.

5.4 Procedure

Participants provided informed consent and completed a brief demographic questionnaire in the beginning. They then received instructions on the experimental controls, specifically how to use the physical dial to transition between worlds and the controller buttons for locomotion within the virtual environment. A short training session followed to familiarize participants with the operation and transition mechanics of all three conditions. To minimize order effects, the sequence of the three transition conditions was counterbalanced across participants using a Latin Square design.

During each condition, participants were instructed to actively explore the virtual environment and perform the photo-taking task described previously. System data, including captured photographs, locomotion trajectories inside VR, and HMD head-tracking data, were automatically logged. To ensure safety, the onset of simulator sickness was continuously monitored using the Fast Motion Sickness scale through periodic verbal checks. No participants reported severe symptoms or required a break during the sessions.

After completing each condition, participants responded to a set of questionnaires that used 7-point Likert scale:

S1 A custom questionnaire on Explore and Control
S2 Presence Questionnaire (PQ) [57], used without additional sections on sound and haptics.
S3 Spatial Presence Experience Scale short version (SPES) [20].
S4 A questionnaire on Transition Continuity adapted from Husung *et al.* [22].
S5 User Experience Questionnaire short version (UEQ-S) [29, 45].

The custom questionnaire on Explore and Control contained the following items:

Explore

1. I feel like I want to explore the virtual environment.
2. I feel like there are many states that produce a useful space / many possibilities to utilize the space.
3. I feel like there are many spaces I would like to use in the virtual environment.

Control

4. I feel that I can control the relationship between myself and the virtual worlds.
5. I feel that I can control my relationship with the virtual objects.
6. I feel that I am in control of where I am within the virtual worlds.

The Explore scale reflects exploratory motivation and perceived affordance, and the Control scale reflects perceived agency and spatial control, customized to fit our frequency-based environmental modulation.

Subsequently, a semi-structured interview was conducted. To facilitate recall and deeper reflection, we started the interview by asking participants to talk about their photography, and based on that talked about their spatial experience. Questions focused on usability, perceived continuity, sense of agency, spatial understanding, and overall preference. The entire procedure lasted approximately 75 to 100 minutes.

5.5 Data Analysis

We analyzed the quantitative data collected from the questionnaires and logged metrics. Given the relatively small sample size ($N = 13$) and the potential difficulty in reliably assessing distributional assumptions (e.g., normality) with such a sample, we applied non-parametric statistical analyses for all variables. We performed the Friedman test to evaluate the main effects of the transition methods (*Fade*, *Dissolve*, and *Spectrum*). We report Kendall's W as the estimate of the effect size for the Friedman test. When the Friedman test revealed significant main effects, we conducted post-hoc pairwise comparisons using Holm-corrected Wilcoxon signed-rank tests. For these pairwise comparisons, we calculated the effect size r ($r = Z/\sqrt{N}$). Following Cohen's guidelines [10], we interpret these effect sizes as small ($r > .10$), medium ($r > .30$), or large ($r > .50$). The statistical significance level was set at $\alpha = .05$.

To contextualize the behavioral and questionnaire results, we performed a reflexive thematic analysis of the semi-structured interviews following established guidelines [9] and Braun and Clarke's 2021 framework. Two authors independently reviewed and coded all transcripts, refined a shared coding structure, and applied it across the dataset.

6 RESULTS AND FINDINGS

6.1 Behavior analysis

We analyzed the behavior data collected during the experiment trials. Where (x, y) denote the participant's position in the Unity world coordinate system (origin at $(0, 0)$).

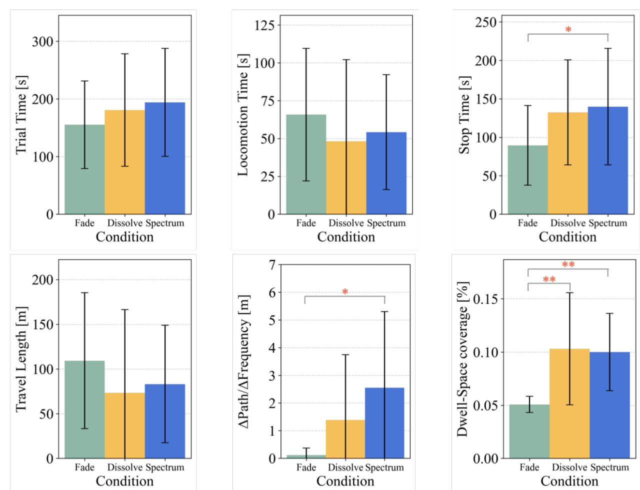


Figure 5: Behavior metrics between conditions: The height of each bar in the graph represents the mean value of the metric under each condition, and the error bars indicate the standard deviation. $^{*}\ p < .05$, $^{**}\ p < .01$.

Locomotion and Temporal Metrics. First, we analyzed basic exploratory behaviors: *Trial Time* and *Travel Length*, calculated as the sum of 2D Euclidean distances ($\sum \sqrt{dx^2 + dy^2}$) in the parallel place. While no statistically significant differences were found across conditions ($p > .05$), the *Spectrum* condition exhibited a descriptive trend toward the longest trial time. Subsequently, we decomposed the trial time into *Locomotion Time* (cumulative time spent moving) and *Stop Time* (cumulative time spent stationary). While no significant difference was found for *Locomotion Time*, a significant main effect was confirmed for *Stop Time* ($\chi^2(2) = 7.54, p < .05, W = 0.29$). Post-hoc analysis revealed that the *Spectrum* condition induced significantly longer stop time compared to the *Fade* condition ($p < .05, r = 0.69$). This result suggests that the continuous spectral transition method encouraged users to stop and observe the environment more frequently.

Interaction and Diversity Metrics. A significant main effect was observed for $\Delta\text{Path}/\Delta\text{Frequency} = \Delta\sqrt{dx^2 + dy^2}/\Delta f$, defined as the distance traveled while the frequency parameter was actively changing ($\chi^2(2) = 11.65, p < .01, W = 0.45$). Specifically, the

Spectrum condition showed significantly higher values compared to *Fade* ($p < .05, r = 0.75$). This indicates that the *Spectrum* condition facilitated a parallel strategy of tuning while moving.

Furthermore, to evaluate the comprehensiveness of exploration, we examined the *Dwell-Space Coverage* metric, calculated as the ratio of frequency bins (0–10000 Hz) where the participant stayed for at least 3 seconds. The threshold of 3 seconds was chosen as an operational criterion to distinguish intentional exploration from transient transitions between frequency bins, based on pilot observations. A strong main effect was confirmed ($\chi^2(2) = 16.20, p < .001, W = 0.62$). Post-hoc tests showed that both *Spectrum* ($p < .01, r = 0.85$) and *Dissolve* ($p < .01, r = 0.88$) resulted in significantly higher diversity compared to *Fade*. This difference is expected given the discrete nature of the spatial transition in the *Fade* condition.

6.2 Questionnaire

We analyzed the subjective feedback collected through the five sections of the questionnaire.

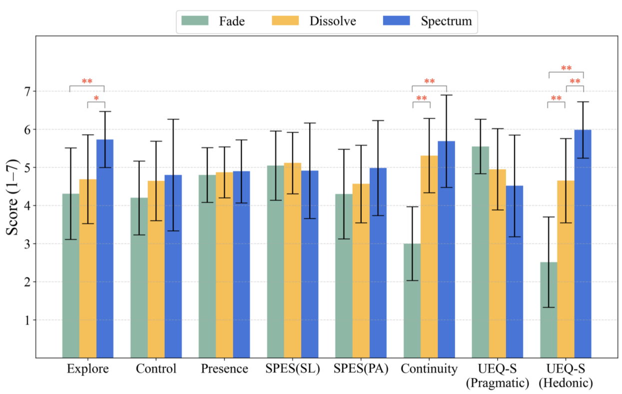


Figure 6: Questionnaire scores: The height of each bar in the graph represents the mean value of the metric under each condition, and the error bars indicate the standard deviation. * $p < .05$, ** $p < .01$.

Exploration and Control (S1). Section 1 comprised six custom items designed to assess the user's exploratory space use, categorizing it into two subscales, *Exploration* and *Control* as described in the above section. Statistical analysis revealed a significant main effect for the total score ($\chi^2(2) = 13.00, p < .01, W = 0.40$). Upon examining the subscales, a significant main effect was found specifically for *Exploration* ($\chi^2(2) = 12.10, p < .01, W = 0.39$). Post-hoc comparisons using Holm's correction indicated that the *Spectrum* condition scored significantly higher than both the *Fade* ($p < .01$) and *Dissolve* ($p < .05$) conditions. Conversely, no significant main effect was observed for *Control* ($\chi^2(2) = 4.35, p = .11, W = 0.14$). These findings suggest that while the proposed method maintains a level of basic controllability comparable to existing techniques, it significantly enhances the user's perception of active exploration.

Presence (S2). Section 2 utilized the Presence Questionnaire (PQ) adapted from Witmer and Singer [57]. The analysis yielded no significant differences between conditions for the total score or across any subscales, including *Realism* ($\chi^2(2) = 1.46, p = .48, W = 0.05$) and *Possibility to Act* ($\chi^2(2) = 0.71, p = .70, W = 0.02$). This implies that the experimental manipulation of transition techniques did not negatively impact the baseline sense of presence.

Spatial Presence Experience Scale (S3). Section 3 employed the short Spatial Presence Experience Scale (SPES) [20], focusing on *Self Location* and *Possible Action*, corresponding to spatial presence and agency. No significant differences were found in the overall SPES score or the *Self Location* subscale. Although the Friedman test detected a significant main effect for the possible action subscale ($\chi^2(2) = 6.14, p < .05, W = 0.19$), subsequent post-hoc analysis with Holm's correction failed to reveal any significant differences between specific pairs (Spectrum vs. Fade: $p = .06$). These results indicate that the proposed method preserves the user's sense of immersion and reality at a level equivalent to established transition methods.

Transition Continuity (S4). Section 4 evaluated continuity using a custom three-item scale adapted from Husung et al. [22], focusing on the perceived smoothness of scene transitions. A robust main effect was confirmed for transition continuity ($\chi^2(2) = 15.97, p < .001, W = 0.52$). Post-hoc tests demonstrated that both the *Spectrum* ($p < .01$) and *Dissolve* ($p < .01$) conditions received significantly higher ratings than the *Fade* condition. No significant difference was observed between the *Spectrum* and *Dissolve* conditions ($p = .35$), indicating that the proposed method achieved a comparable level of perceptual continuity to the conventional *Dissolve* technique.

User Experience (S5). The User Experience Questionnaire (UEQ-Short) [45] presented contrasting outcomes for its two underlying dimensions: *Pragmatic Quality* and *Hedonic Quality*. While no significant difference was found for *Pragmatic Quality* (usability) ($\chi^2(2) = 4.77, p = .09, W = 0.14$), a substantial main effect was confirmed for *Hedonic Quality* (stimulation/novelty) with a large effect size ($\chi^2(2) = 26.53, p < .001, W = 0.88$). Post-hoc comparisons revealed that the *Spectrum* condition was rated significantly higher than both *Dissolve* ($p < .01$) and *Fade* ($p < .01$). Additionally, *Dissolve* was rated significantly higher than *Fade* ($p < .01$). This dichotomy suggests that the interaction of "tuning frequencies to discover worlds" delivers exceptional experiential value—characterized by enjoyment and novelty—without compromising pragmatic usability.

6.3 Interview

Below, we present the resulting themes with representative quotations and their links to each condition.

6.3.1 *Theme 1: Adaptation and Active Use of the System.*

Although *Spectrum* condition was unfamiliar compared to conventional methods, participants described this uncertainty as engaging.

P3: "*Surprise, not in a negative sense, but in the sense of 'Ah, so this is how it unfolds.'*"

P14: "*I lean toward curiosity...this motivated me to explore more particularly with the spectrum condition.*"

Participants quickly formed their own interpretations of the experience, which can sometimes differ across individuals.

P6: "*the player essentially moves along this frequency range and encounters different planes of reality.*"

P2:"*The worlds were congruent in geometry...I felt like I was interpolating through the objects.*"

P1: "*I co-exist in multiple worlds.*"

They also actively experimented with the system to explore the virtual worlds and create specific photographic compositions.

P15: "*With frequency transitions, you can't really predict what will appear or disappear... I wanted to use the differences between the environments to include the red flowers, the tent on this side, and the forest-like part all at the same time.*"

P7: (referring to Fig. 7(a))"*I enjoy blending scenes because these different worlds have distinct characteristics and atmospheres... so I adjust the controls to include elements from multiple environments.* "

6.3.2 ***Theme 2: Spatial Usability of Transitional States****.* Participants consistently described the *Spectrum* condition as producing usable intermediate spaces rather than mere transitions. Most participants photographed these in-between states because they perceived them as a sequence of layered, functional realities.

P2:"*Every step that I took was a completely usable space… All the degrees in the frequency were somehow usable.*"

P9: (referring to Fig. 7(b))"*All states, including the intermediate ones, felt like complete worlds.*"

While the *Dissolve* condition provided continuity, the in-between states were experienced as non-functioning.

P2: "*The dissolve condition feels like a liminal space somewhere in-between where I cannot really exist. When I was between spaces it did not feel like a usable space.*"

P9: "*During the transition I don't perceive it as a space; it's merely a passageway. Since I don't linger there, I didn't need to photograph it.*"

6.3.3 ***Theme 3: Self-Location During Transition****.* Participants' exploratory engagement depended strongly on how each transition method shaped their sense of self-location. The *Spectrum* condition offered the strongest feeling of being anchored in place, which is perceived as useful.

P5: "*felt less as though I was moving myself and more as though the world was approaching me,*"

P11: "*I simply remain where I was while the world changes around me..*"

P6: "*what is quite practical about Spectrum condition is that you realize where you are standing in each scene, and you know where you're gonna land when you transition fully. With Fade condition, I suddenly was outside of the office, even though I wanted to be inside, because I moved in the forest.*"

While the *Dissolve* condition uses similar mechanism, it was described as destabilizing, leading to a sense of being caught in an indeterminate state.

P11: "*losing track of which level I am in.*"

P2: "*interpolating the entire world in a way that obscured the relationship to objects and layers.*"

6.3.4 ***Theme 4: Interpreting Spatial Structure Through Layered Transitions****.* Participants described the *Spectrum* condition as providing layered information about both individual objects and the broader environment, making the transition process itself a perceptual aid.

For object-level structure, spectrum design offers different stages and levels of detail of one object.

P4: (referring to Fig. 7(c)) "*It's not simply becoming more transparent—I think you can also see the internal structure in this way.*"

P2: (referring to Fig. 7(d))"*You can also remove some details from the same object... at this point I just end up with the frame of the totem, and I thought it was interesting—like a glass box—to look inside and use the frames for the photo.*"

For environment-level structure, *Spectrum* condition also clarified scene composition by introducing elements incrementally.

P11:"*The frequency method introduces things one at a time or in phases creating more interesting combinations whereas with dissolve everything happens at once. The dissolve effect brings everything through simultaneously which can be visually overwhelming.*"

6.3.5 ***Theme 5: Enjoyment****.* Participants ranked the three transition methods in order of preference. *Spectrum* condition received the highest ranking 13 times (with an additional 3 ties with *Dissolve* condition), *Dissolve* was ranked first 4 times (with 3 ties), and *Fade* was ranked first 1 time. Several participants attributed their preference to the enjoyment of the spatial experience itself.

P11: "*The spectrum condition was more enjoyable and interesting not being entirely certain what would appear next. Since photography is inherently creative and exploratory this technique complemented the task well, making it fun and engaging. The uncertainty of not knowing exactly which frequency produces which elements added to the exploratory nature.*"

P10: "*As I turned the knob I was surprised by the transitional visual. It felt funky and interesting somewhat trippy so I photographed it.*"

7 DISCUSSION

We conducted this study to evaluate the designed system, and investigate how people's perceptions and behaviours change. The following discusses these questions and limitations for future work.

7.1 Usable States Between Discrete Scenes

Both the *Spectrum* and *Dissolve* conditions produce visually continuous intermediate states as reflected in the *Transition Continuity* results, where these conditions were both rated significantly higher than *Fade*, and showed no significant difference between each other. However, their *spatial function* differs markedly. Results of the Explore subscale in S1 showed that the *Spectrum* condition was rated significantly higher than both *Fade* and *Dissolve*, indicating users perceive more "spaces to use" and higher "will to explore". Interview findings (Theme 2) directly support this: Spectrum's in-between states were consistently described as *usable worlds*, and Interview findings (Theme 4) depict the functions of the in-between spaces to provide additional levels of information. Behavioral metrics align with these perceptions. Although overall trial duration showed no significant effect, the Spectrum condition exhibited longer stop time and a trend toward longer exploration, indicating that users not only perceived these intermediate states as usable but actively engaged with them.

(a) (b) (c) (d)

Figure 7: Participant photographs captured during the experiment under Condition 3. (a) creating through exploration, (b) usable in-between states, (c) observing internal structure, (d) partially visible objects supporting creativity.

Together, these results show that Spectral Space enhances the virtual environment by generating meaningful, usable states between otherwise discrete scenes. It generates a much wider framework for spatial construction by adding only one non-spatial, one-dimensional axis. This new design space can be particularly valuable for layered game design, in both entertainment and education, or for novel co-presence spaces in multi-user scenarios. In scenarios when a fast, discrete transition is required, traditional methods like *Fade* can be preferred, as one participant ranked *Fade* as their favourite method as they are effectiveness-oriented.

7.2 Changes in Users' Exploratory Behaviour and Strategy

The strongest effects of the *Spectrum* condition appeared in measures related to user exploration. Subjectively, participants reported significantly greater *Exploration* and substantially higher Hedonic Quality compared to both control conditions. Behavioral data agree with these self-reports and reveal a distinct exploratory strategy compared to the conventional *Fade* condition. Significantly higher $\Delta Path/\Delta Frequency$ shows users are adopting a parallel strategy of "tuning while moving." This integration of locomotion and parameter adjustment is characteristic of the *Spectrum* approach and likely contributes to its elevated hedonic evaluations. Although *Fade* showed a descriptive tendency toward greater path length and locomotion time, the *Spectrum* condition elicited significantly longer Stop Time, indicating that users paused deliberately to observe unfolding spatial layers rather than transitioning quickly. Interview data further supports this pattern. Participants described the spectral transition as motivating curiosity and experimentation (Theme 1) and emphasized the enjoyment derived from uncertainty, surprise, and the creative potential of layered transitions (Theme 5).

Together, these findings show the spectrum design mechanism encourages a more exploratory, playful, and strategically engaged spatial awareness. The "tuning while moving" behavior supports open-ended interaction, making it well-suited for creative tools, immersive exhibitions, or reflective learning spaces. Rather than guiding users along predefined paths, Spectrum-like systems may invite users to develop personal spatial narratives, enabling more human-centered forms of immersion where meaning unfolds through interaction.

7.3 Perceptual Change With Stable Presence

We find that Spectral Space supports a dual effect: it preserves a stable sense of presence while simultaneously enabling a transformed perceptual relationship to surrounding worlds. Across conditions, the *Spectrum* method maintained users' baseline presence and self-location—S2 and S3 showed no significant differences—demonstrating that adding a non-spatial frequency axis does not disrupt the fundamental feeling of "being there."

At the same time, participants reported a distinctly different perceptual experience. They described the spectral transition as moving *through* multiple worlds while remaining physically anchored—observing layered scene changes without any shift in self-location—as reflected in Theme 1 and Theme 3. Furthermore, these layered transitions revealed structural information about objects and environments (Theme 4) that reshaped how participants interpreted each world. This stands in contrast to conventional multi-presence, which typically relies on attention switching or discrete scene transitions rather than inhabiting a continuous, structurally informative perceptual spectrum.

Although novel, the mechanism did not introduce cognitive burden. As Theme 1 shows, initial surprise quickly gave way to coherent mental models and active use of the frequency axis for exploring object layers and composing photographs. This adaptability is supported by questionnaire S5: high Hedonic Quality and unchanged Fragmentation scores indicate that the method was stimulating rather than confusing.

Together, these results show that users can reinterpret and traverse multi-layered space dynamically while remaining anchored. This suggests that spectral transitions support a deeper form of presence—not defined by spatial certainty, but by perceptual participation. The experience becomes less about being placed into a world, and more about actively composing it through subtle, embodied negotiation. In this way, the system expands how users can interpret, inhabit, and personalize immersive space.

7.4 Limitations

This study has several limitations. First, the sample size was small and predominantly young male, limiting generalizability. Second, the task focused on short-term exploratory observation within predefined virtual worlds, and did not address longer-term or task-oriented use. Third, the semantic mapping of frequency was intentionally minimal, which led to varied interpretations among participants and may have influenced subjective responses. Finally, object manipulation and multi-user interaction were not evaluated, leaving their role in more complex or collaborative scenarios unexplored. These limitations indicate directions for future work rather than fundamental issues with the framework itself.

7.5 Future Work

7.5.1 Co-Presence and Interaction Among Users. Although Spectral Space supports multi-user interaction, it was not formally evaluated. Preliminary workshop suggests that frequency tuning enables novel forms of social navigation, such as selectively aligning with others for interaction or maintaining partial misalignment for peripheral awareness. Future work should examine how spectral alignment influences collaboration, coordination, privacy, and shared awareness in multi-user environments.

7.5.2 Potential in Modality Extensions. Currently, frequency is adjusted manually via a dial and affects only visual content. Future research could explore alternative input mappings, such as physiological signals (e.g., heart rate or affect), enabling adaptive or socially synchronized environments. Extending spectral modulation to additional modalities—such as spatial audio or haptics—may further enrich layered perception and strengthen the integration of bodily signals with virtual space.

7.5.3 Interaction With Objects as a Design Toolkit. Preliminary support for carrying objects across frequencies suggests that Spectral Space could function as a design toolkit. By tuning frequency rather than relocating spatially, users can compose hybrid or intermediate spaces within a shared layout. Future work should investigate how such object-level control supports creative authorship, customization, and exploratory spatial design in VR.

7.6 CONCLUSION

This paper introduced *Spectral Space*, a framework that organizes virtual environments along a continuous non-spatial frequency axis rather than as discrete scenes. Through a system implementation and comparative user study, we showed that spectral modulation creates usable intermediate spaces, supports layered interpretation of objects and environments, and encourages more exploratory behavior in virtual space. Despite this unconventional spatial structure, users maintained stable presence and self-location while benefiting from a more flexible and expressive spatial experience. These findings suggest that treating virtual worlds as continuous spectral layers can expand the design space of VR, offering new possibilities for interaction, navigation, and multi-environment coherence. Future work will explore multi-user spectral co-presence, links between spectral transitions and somatic signals, and the use of Spectral Space as a design toolkit.

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